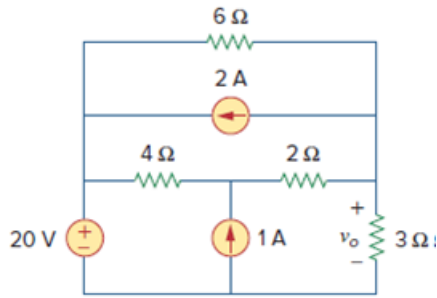


Problem

Apply the superposition principle to find v_o in the circuit of Fig. 4.82.

Figure 4.82:



Step-by-step solution

Step 1 of 9

Refer to figure 4.82 given in the textbook.

Apply Superposition principle to the circuit.

Consider a single source among three sources and obtain the values of output voltage v_{o1} , v_{o2} , and v_{o3} individually.

$$v_o = v_{o1} + v_{o2} + v_{o3} \dots\dots (1)$$

Step 2 of 9

Consider the voltage source of 20 V and open circuit the current sources of 1 A and 2 A as shown in figure 1.

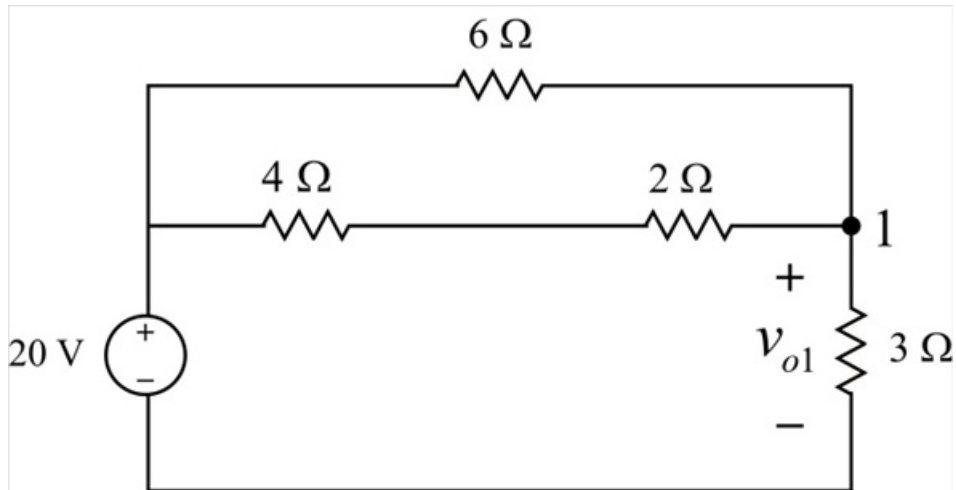


Figure 1

Step 3 of 9

Apply Nodal analysis at node-1.

$$\frac{v_{o1} - 20}{6} + \frac{v_{o1}}{3} + \frac{v_{o1} - 20}{6} = 0$$

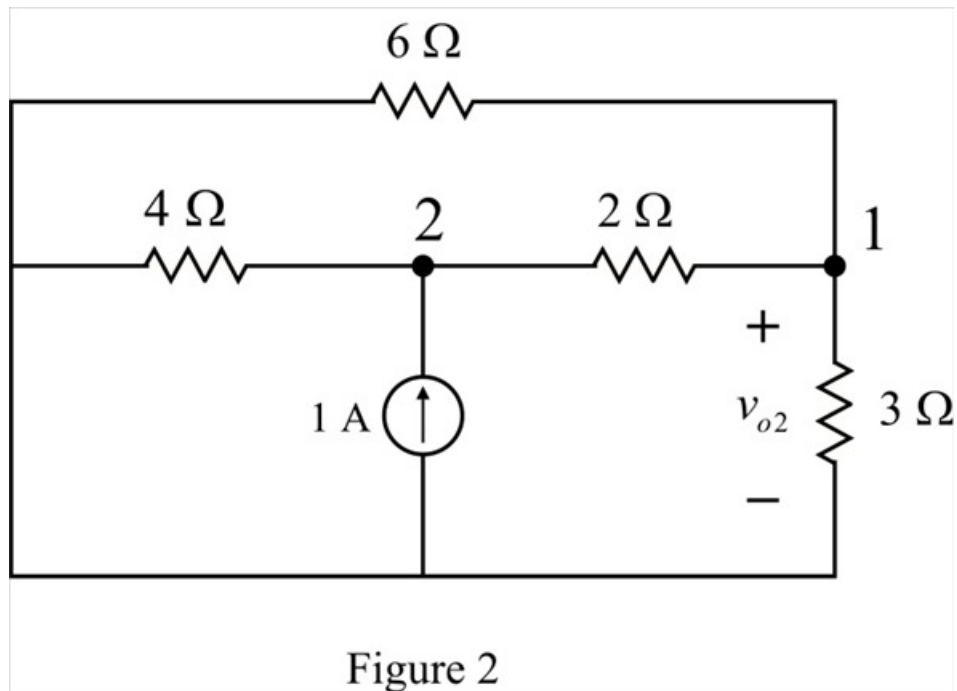
$$v_{o1} \left(\frac{1}{6} + \frac{1}{6} + \frac{1}{3} \right) = \frac{20}{6} + \frac{20}{6}$$

$$v_{o1} \left(\frac{2}{3} \right) = \frac{20}{3}$$

$$v_{o1} = 10 \text{ V}$$

Step 4 of 9

Consider the current source of 1 A, open circuit the current source of 2 A and short circuit the voltage source of 20 V as shown in figure 2.



Step 5 of 9

Assume that the voltage of node-2 as v_1 .

Apply Nodal analysis at node2.

$$-1 + \frac{v_1}{4} + \frac{v_1 - v_{o2}}{2} = 0$$

$$v_1 \left(\frac{1}{2} + \frac{1}{4} \right) = 1 + \frac{v_{o2}}{2}$$

$$v_1 = \frac{4}{3} \left(1 + \frac{v_{o2}}{2} \right) \dots\dots (2)$$

Step 6 of 9

Apply Nodal analysis at node-1.

$$\frac{v_{o2}}{3} + \frac{v_{o2}}{6} + \frac{v_{o2} - v_1}{2} = 0$$

$$\frac{v_{o2}}{3} + \frac{v_{o2}}{6} - \left(\frac{v_1 - v_{o2}}{2} \right) = 0$$

$$v_{o2} \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{2} \right) = \frac{v_1}{2}$$

$$v_{o2} = \frac{v_1}{2}$$

Consider equation (2). Substitute $\frac{4}{3} \left(1 + \frac{v_{o2}}{2} \right)$ for v_1 .

$$v_{o2} = \frac{1}{2} \times \frac{4}{3} \left(1 + \frac{v_{o2}}{2} \right)$$

$$v_{o2} = \frac{2}{3} + \frac{v_{o2}}{3}$$

$$\frac{2}{3} v_{o2} = \frac{2}{3}$$

$$v_{o2} = 1 \text{ V}$$

Step 7 of 9

Consider the current source of 2 A, open circuit the current source of 1 A and short circuit the voltage source of 20 V as shown in figure 3.

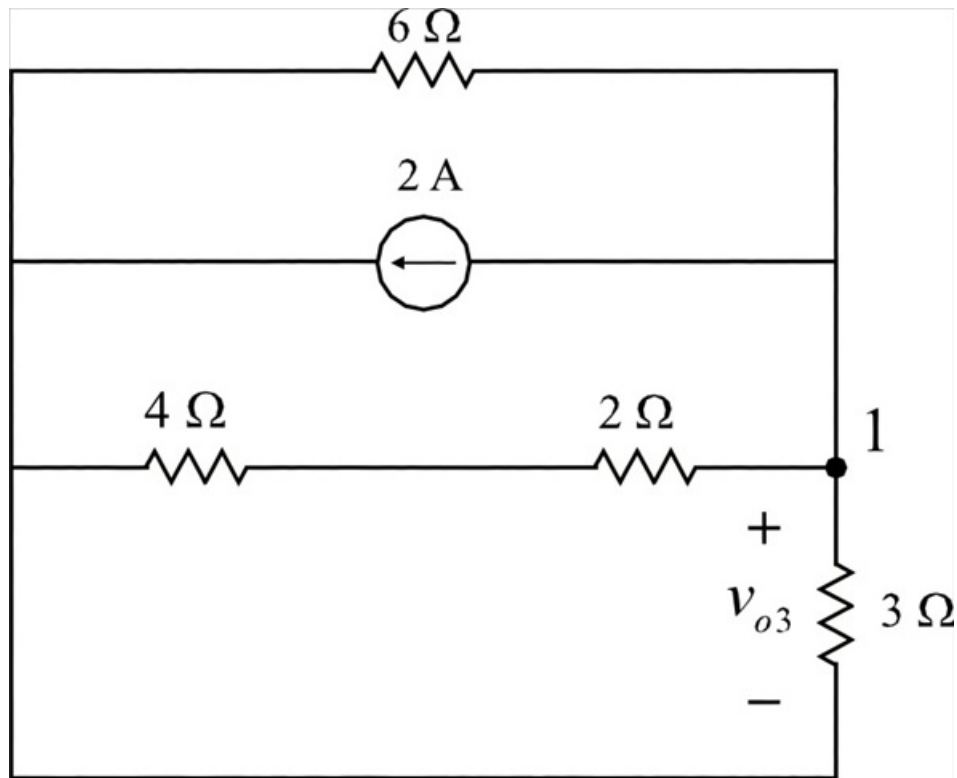


Figure 3

Step 8 of 9

Apply Nodal analysis at node-1.

$$\frac{v_{o3}}{3} + \frac{v_{o3}}{6} + 2 + \frac{v_{o3}}{6} = 0$$

$$v_{o3} \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{6} \right) = -2$$

$$v_{o3} \left(\frac{2}{3} \right) = -2$$

$$v_{o3} = -3 \text{ V}$$

Step 9 of 9

Recall equation (1).

$$v_o = v_{o1} + v_{o2} + v_{o3}$$

Substitute 10 for v_{o1} , 1 for v_{o2} , and $\diamond\diamond$ for v_{o3} .

$$\begin{aligned} v_o &= 10 + 1 - 3 \\ &= 8 \text{ V} \end{aligned}$$

Therefore, the value of voltage v_o is 8 V.